Reconstruction from Observation:
The Density Bridge and Measure-Theoretic Takens

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Abstract

Given the probability measure produced by Paper I and the dynamics of Paper II, this paper asks when an observer can recover the full state space from time-delayed measurements alone. The answer is the *reconstruction theorem*: for a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and a bounded observation function $h \in L^\infty(X, \mu)$, the following are equivalent: (i) the observable σ-algebra $\mathcal{O}_h = \sigma(\{h \circ T^n : n \in \mathbb{Z}\})$ equals $\mathcal{B}(X)$ modulo μ; (ii) the algebra generated by the delay observations $\{h \circ T^n\}$ is dense in $L^2(X, \mu)$; (iii) the delay map $\Phi_h : x \mapsto (h(T^n x))_{n \in \mathbb{Z}}$ is a measure-theoretic embedding $X \hookrightarrow \mathbb{R}^{\mathbb{Z}}$.

The equivalence of (i) and (ii) is the *density bridge*: the L^2-closure of an algebra of bounded measurable functions equals the L^2 space of the σ-algebra it generates. This is proved via the functional monotone class theorem and simple-function approximation.

The cyclic vector condition from Paper II — that $\overline{\text{span}}\{U_T^n h\} = L^2(X, \mu)$ — implies (i)–(iii) but is strictly stronger: it is a spectral condition on U_T, while reconstruction is a condition on the Boolean algebra of observable events. The two coincide when U_T has simple L^2-spectrum.

The reconstruction theorem closes the loop with Paper I: the Stone space $\text{St}(\mathcal{O}_h)$ constructed there as a technical tool for measure extension is, under condition (i), measure-theoretically isomorphic to X itself. The programme begins with a compact completion of the sample space and ends by identifying it with the state space.

Compared to the classical Takens embedding theorem [5], the present result requires only measurability, uses all finite delays rather than a fixed window, and replaces the dimension count with the algebraic density condition.

1 Introduction

The problem. Papers I and II established that coherent structured observations determine a probability measure, and that measure determines the dynamics. The remaining question is whether the observer can recover the state space itself. Given only the time series $h(x), h(Tx), h(T^2x), \ldots$, can they reconstruct where in X they are?

The answer depends on how much information the observation function h carries about the state. If h is constant, no reconstruction is possible. If the orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ collectively separates almost every pair of points, reconstruction holds. The reconstruction theorem makes this precise.

The density bridge. The central result is that two conditions — one on the Boolean algebra of observable events, one on the L^2-approximation theory of the delay observations — are equivalent. The bridge between them is a general lemma on function algebras: for a subalgebra $\mathcal{A} \subseteq L^\infty(X, \mu)$, the L^2-closure of $\mathcal{A}$ equals $L^2(X, \sigma(\mathcal{A}), \mu)$. This is an instance of the functional monotone class theorem.

The cyclic vector condition. The Koopman operator U_T on $L^2(X, \mu)$ provides a spectral perspective. The condition that h is a *cyclic vector* for U_T — $\overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2$ — implies reconstruction. The implication is strict: reconstruction asks only that the *algebra* generated by $\{h \circ T^n\}$ is L^2-dense, while cyclicity asks that the *linear span* is dense. These coincide if and only if U_T has simple L^2-spectrum.

The Stone space connection. Paper I [4] built the Stone space $\mathrm{St}(\mathcal{C})$ as a compact Hausdorff completion of the sample space for the purpose of measure extension. Paper III identifies the Stone space of the observable algebra $\mathcal{O}_h$ with the state space X when reconstruction holds. The compact space introduced as a technical device in Paper I reappears here as the object being recovered.

Relation to Takens. The classical Takens theorem [5] requires a smooth vector field on a compact d-manifold, a generic observation function, and $2d+1$ delays; it produces a diffeomorphic embedding. The reconstruction theorem here requires only measurability, uses all finite delays, and replaces the dimension count with the algebraic density condition. On a compact manifold, the density condition holds generically, recovering the spirit of Takens without differentiability hypotheses.

Organisation. Section 2 sets up the dynamical system, observation function, and observable algebra. Section 3 proves the density bridge lemma. Section 4 states and proves the reconstruction theorem. Section 5 places the cyclic vector condition in context. Section 6 closes the loop with Paper I. Section 7 compares with the classical Takens theorem.

2 Setup

Throughout this paper, $(X, \mathcal{B}, \mu)$ is a standard probability space and $T : X \to X$ is a measurable, μ-preserving map. We assume T is *invertible*: T is a bimeasurable bijection with $T_*\mu = \mu$ and $(T^{-1})_*\mu = \mu$.

We fix a bounded observation function $h \in L^\infty(X, \mu)$. The invertibility of T allows us to use both forward and backward delays:

Definition 2.1 (Observable algebra)**.** The *observable algebra* generated by h is

$$\mathcal{O}_h \;:=\; \sigma\big(\{h \circ T^n : n \in \mathbb{Z}\}\big) \;\subseteq\; \mathcal{B}(X).$$

The *delay algebra* is the algebra of finite polynomials in the delay observations:

$$\mathcal{A}_h \;:=\; \mathrm{alg}\big(\{h \circ T^n : n \in \mathbb{Z}\} \cup \{1\}\big) \;\subseteq\; L^\infty(X, \mu),$$

the smallest subalgebra of $L^\infty(X, \mu)$ containing 1 and every $h \circ T^n$. Concretely, $\mathcal{A}_h$ consists of all finite sums $\sum_\alpha c_\alpha \prod_k (h \circ T^{n_k})^{e_k}$ with $c_\alpha \in \mathbb{R}$, $n_k \in \mathbb{Z}$, $e_k \in \mathbb{N}$.

Definition 2.2 (Delay map)**.** The *delay map* is

$$\Phi_h : X \to \mathbb{R}^{\mathbb{Z}}, \qquad \Phi_h(x) \;=\; \big(h(T^n x)\big)_{n \in \mathbb{Z}}.$$

Since each coordinate $x \mapsto h(T^n x)$ is measurable and $\mathbb{R}^{\mathbb{Z}}$ carries the product Borel σ-algebra $\mathcal{B}(\mathbb{R}^{\mathbb{Z}}) = \sigma(\{\pi_n : n \in \mathbb{Z}\})$, the map Φ_h is measurable.

Lemma 2.3 (Observable algebra as pullback)**.** *$\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$ mod μ.*

Proof. The product σ-algebra $\mathcal{B}(\mathbb{R}^{\mathbb{Z}})$ is generated by coordinate projections $\{\pi_n : n \in \mathbb{Z}\}$. Since $\pi_n \circ \Phi_h = h \circ T^n$, the pullback $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}}))$ is generated by $\{(h \circ T^n)^{-1}(E) : E \in \mathcal{B}(\mathbb{R}),\, n \in \mathbb{Z}\}$, which is exactly $\mathcal{O}_h$. □

The observation that $\mathcal{O}_h$ is the pullback σ-algebra of Φ_h means that $\mathcal{O}_h = \mathcal{B}$ mod μ is precisely the condition that Φ_h is a measure-theoretic embedding — condition (iii) of the reconstruction theorem. The content of the theorem is that this geometric condition is equivalent to the analytic condition (ii) on algebra density.

3 The density bridge

The density bridge is the lemma connecting the Boolean algebra level (the σ-algebra $\mathcal{O}_h$) to the L^2 level (approximation by elements of $\mathcal{A}_h$). It holds in complete generality for any subalgebra of L^∞.

Lemma 3.1 (Density bridge)**.** *Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subseteq L^\infty(X, \mu)$ be a subalgebra containing the constant function 1. Then:*

$$\overline{\mathcal{A}}^{L^2} = L^2(X, \sigma(\mathcal{A}), \mu).$$

Consequently, $\overline{\mathcal{A}}^{L^2} = L^2(X, \mu)$ if and only if $\sigma(\mathcal{A}) = \mathcal{B}$ modulo μ.

Proof. Write $\mathcal{M} := \sigma(\mathcal{A})$ and $V := \overline{\mathcal{A}}^{L^2}$.

Step 1: $V \subseteq L^2(X, \mathcal{M}, \mu)$**.** Every $f \in \mathcal{A}$ is $\mathcal{A}$-measurable, hence $\mathcal{M}$-measurable. So $\mathcal{A} \subseteq L^2(X, \mathcal{M}, \mu)$. Since $L^2(X, \mathcal{M}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$, taking the closure gives $V \subseteq L^2(X, \mathcal{M}, \mu)$.

Step 2: $L^2(X, \mathcal{M}, \mu) \subseteq V$**.** It suffices to show that every indicator $\mathbf{1}_E$ with $E \in \mathcal{M}$ lies in V. By the functional monotone class theorem [1], the class

$$\mathcal{H} := \{f \in L^\infty(X, \mathcal{M}, \mu) : f \in V\}$$

is a vector space containing $\mathcal{A}$ and closed under bounded monotone pointwise limits (since if $f_n \in V$ are uniformly bounded and $f_n \to f$ pointwise a.e. with $f \in L^\infty$, then $f_n \to f$ in L^2 by dominated convergence, so $f \in V$). Since $\mathcal{A}$ is an algebra containing 1 and $\sigma(\mathcal{A}) = \mathcal{M}$, the monotone class theorem gives $\mathcal{H} = L^\infty(X, \mathcal{M}, \mu)$. In particular every $\mathcal{M}$-indicator $\mathbf{1}_E \in V$.

Since the $\mathcal{M}$-simple functions are dense in $L^2(X, \mathcal{M}, \mu)$ and V is closed, $L^2(X, \mathcal{M}, \mu) \subseteq V$.

Conclusion. Steps 1 and 2 give $V = L^2(X, \mathcal{M}, \mu)$. The final equivalence is immediate: $V = L^2(X, \mathcal{B}, \mu)$ iff $L^2(X, \mathcal{M}, \mu) = L^2(X, \mathcal{B}, \mu)$ iff $\mathcal{M} = \mathcal{B}$ mod μ. □

Remark 3.2 (Role of the algebra structure)**.** The algebra structure of $\mathcal{A}$ is essential. The corresponding statement for the closed *linear* span $\mathcal{H}_\mathcal{A} = \overline{\operatorname{span}}(\mathcal{A})$ is false in general: $\mathcal{H}_\mathcal{A} = L^2$ does not imply $\sigma(\mathcal{A}) = \mathcal{B}$.

For example, take $X = [0, 1]$, $\mu =$ Lebesgue, $T(x) = 2x \bmod 1$ (the doubling map), and $h(x) = x$. The orbit $\{h \circ T^n\}(x) = \{2^n x \bmod 1\}$ generates $\mathcal{B}([0, 1])$ as a σ-algebra (the dyadic structure separates points). But $\operatorname{span}\{2^n x \bmod 1 : n \geq 0\}$ is a proper subspace of $L^2[0, 1]$: it does not contain x^2. So the linear span can be a proper dense subspace even when $\sigma(\mathcal{A}) = \mathcal{B}$.

In the other direction, h cyclic for U_T (linear span dense) always implies $\sigma(\mathcal{A}_h) = \mathcal{B}$ mod μ — see Corollary 5.2.

4 The reconstruction theorem

Theorem 4.1 (Reconstruction theorem)**.** *Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system and $h \in L^\infty(X, \mu)$. The following are equivalent:*

(i) $\mathcal{O}_h = \mathcal{B}$ modulo μ;

(ii) $\mathcal{A}_h$ is dense in $L^2(X, \mu)$;

(iii) $\Phi_h : X \to \mathbb{R}^\mathbb{Z}$ is a measure-theoretic embedding: $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B}$ modulo μ.

Proof. (i) ⇔ (ii). Apply the density bridge (Lemma 3.1) with $\mathcal{A} = \mathcal{A}_h$ and $\sigma(\mathcal{A}_h) = \mathcal{O}_h$: $\overline{\mathcal{A}_h}^{L^2} = L^2(X, \mu)$ iff $\mathcal{O}_h = \mathcal{B}$ mod μ.

(i) ⇔ (iii). By Lemma 2.3, $\mathcal{O}_h = \Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z}))$ mod μ. So $\mathcal{O}_h = \mathcal{B}$ iff $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^\mathbb{Z})) = \mathcal{B}$ mod μ, which is exactly condition (iii). □

Remark 4.2 (What "measure-theoretic embedding" means)**.** Condition (iii) says that Φ_h is *almost everywhere injective* and the pullback σ-algebra is all of $\mathcal{B}$. More precisely: since $\Phi_h^{-1}(\mathcal{B}(\mathbb{R}^{\mathbb{Z}})) = \mathcal{B} \bmod \mu$, for any two sets $A, A' \in \mathcal{B}$ with $\mu(A \triangle A') > 0$ there exists a Borel set $E \subseteq \mathbb{R}^{\mathbb{Z}}$ with $\mu(\Phi_h^{-1}(E) \triangle A) = 0$ and $\mu(\Phi_h^{-1}(E) \triangle A') > 0$. In measure-theoretic language, Φ_h induces an isomorphism of measure algebras $(X/\mu, \mathcal{B}/\mu) \cong (\mathbb{R}^{\mathbb{Z}}/(\Phi_h)_*\mu, \mathcal{B}(\mathbb{R}^{\mathbb{Z}})/(\Phi_h)_*\mu)$.

The pushforward measure $(\Phi_h)_*\mu$ is concentrated on the image $\Phi_h(X) \subseteq \mathbb{R}^{\mathbb{Z}}$ by definition. The T-invariance of μ implies that $(\Phi_h)_*\mu$ is invariant under the bilateral shift σ on $\mathbb{R}^{\mathbb{Z}}$: the dynamical system (X, T, μ) is isomorphic to $(\mathbb{R}^{\mathbb{Z}}, \sigma, (\Phi_h)_*\mu)$ restricted to the image of Φ_h.

Corollary 4.3 (Shift invariance)**.** *Under condition (i)–(iii), the map Φ_h intertwines T on X with the bilateral shift σ on $\mathbb{R}^{\mathbb{Z}}$: $\Phi_h \circ T = \sigma \circ \Phi_h$ μ-almost everywhere.*

Proof. For each $n \in \mathbb{Z}$, $(\Phi_h(Tx))_n = h(T^n(Tx)) = h(T^{n+1}x) = (\Phi_h(x))_{n+1} = (\sigma(\Phi_h(x)))_n$. □

5 The cyclic vector condition

The Koopman operator U_T on $L^2(X, \mu)$ encodes the dynamics at the operator level. Recall from Paper II [3] that U_T is unitary (since T is invertible and measure-preserving) and that the predictive operators $\{K_t\}$ are related to U_T via the Koopman–Perron duality.

Definition 5.1 (Cyclic vector)**.** The function $h \in L^2(X, \mu)$ is a *cyclic vector* for U_T if

$$\mathcal{H}_h \ := \ \overline{\operatorname{span}}\{U_T^n h : n \in \mathbb{Z}\} \ = \ L^2(X, \mu).$$

Corollary 5.2 (Cyclic implies reconstruction)**.** *If $h \in L^\infty(X, \mu)$ is a cyclic vector for U_T, then $\mathcal{O}_h = \mathcal{B}$ modulo μ.*

Proof. Since $\mathcal{A}_h$ contains $\operatorname{span}\{h \circ T^n\} = \operatorname{span}\{U_T^n h\}$ as a subset, $\overline{\mathcal{A}_h}^{L^2} \supseteq \mathcal{H}_h = L^2$. By the density bridge, $\mathcal{O}_h = \mathcal{B} \bmod \mu$. □

Remark 5.3 (The implication is strict)**.** The converse of Corollary 5.2 fails in general: $\mathcal{O}_h = \mathcal{B}$ does not imply h is cyclic. The doubling map example of Remark 3.2 witnesses this.

The cyclic vector condition asks for density of the *linear span* of the orbit; reconstruction asks only for density of the *algebra*. For a bounded h, the linear span is contained in the algebra, so cyclicity is the stronger condition.

The two conditions coincide precisely when U_T has *simple L^2-spectrum*: $L^2(X, \mu)$ is a cyclic U_T-module, meaning there exists some cyclic vector. In this case, every generating observation h (satisfying $\mathcal{O}_h = \mathcal{B}$) is automatically cyclic. Simple spectrum is a generic property of ergodic transformations in the Baire category sense [2] but is not universal.

Remark 5.4 (Operator-algebraic perspective)**.** The distinction between reconstruction and cyclicity reflects a level in the Stone–Gelfand–Naimark duality:

Level	**Object**	**Reconstruction condition**
Boolean algebra	$\mathcal{O}_h \subseteq \mathcal{B}$	$\mathcal{O}_h = \mathcal{B} \bmod \mu$
L^2 function space	$L^2(\mathcal{O}_h) \subseteq L^2(\mathcal{B})$	$L^2(\mathcal{O}_h) = L^2(\mathcal{B})$
von Neumann algebra	$\{U_T^n h\}'' \subseteq \mathcal{B}(L^2)$	$\{U_T^n h\}'' = L^\infty(X, \mu)$

The reconstruction theorem operates at the first two levels, which are equivalent by the density bridge. The cyclic vector condition operates at the third level. These coincide when $L^\infty(X, \mu)$ is generated as a von Neumann algebra by the orbit of h under U_T *and* h is a cyclic vector — which holds iff U_T has simple spectrum and h is a generator.

6 The Stone space identification

Paper I constructed the Stone space $\mathrm{St}(\mathcal{C})$ of the cylinder algebra $\mathcal{C}$ as the canonical compact Hausdorff completion of the sample space Ω. In the present setting, the cylinder algebra is replaced by the observable algebra $\mathcal{O}_h$, and the Stone space $\mathrm{St}(\mathcal{O}_h)$ is the compact completion of X for the topology generated by observable events.

Proposition 6.1 (Stone space identification)**.** *Suppose condition (i) of Theorem 4.1 holds: $\mathcal{O}_h = \mathcal{B}$ modulo μ. Then the Stone embedding*

$$\iota : X \to \mathrm{St}(\mathcal{O}_h), \qquad \iota(x) = \{E \in \mathcal{O}_h : x \in E\},$$

is a measure-theoretic isomorphism: the pushforward $\iota_*\mu$ is a probability measure on $\mathrm{St}(\mathcal{O}_h)$, and ι identifies the measure algebra $(X/\mu, \mathcal{B}/\mu)$ with $(\mathrm{St}(\mathcal{O}_h)/\iota_*\mu, \mathcal{B}(\mathrm{St}(\mathcal{O}_h))/\iota_*\mu)$.

Proof. When $\mathcal{O}_h = \mathcal{B}$ mod μ, the Stone embedding ι has the following properties.

Injectivity mod μ: suppose $\iota(x) = \iota(x')$, meaning x and x' belong to the same sets in $\mathcal{O}_h$. For any $E \in \mathcal{B}$ with $x \in E \not\ni x'$, the assumption $\mathcal{O}_h = \mathcal{B}$ mod μ gives $F \in \mathcal{O}_h$ with $\mu(E \triangle F) = 0$. If $x \in F$ then $x' \in F$ (since $\iota(x) = \iota(x')$), so $x' \in F \triangle E \subseteq E^c$, forcing $x' \in (E \triangle F) \cup F^c$. This places x' in a set of measure zero. Hence $\{(x, x') : \iota(x) = \iota(x'),\ x \neq x'\}$ is $(\mu \otimes \mu)$-null, so ι is injective μ-a.e.

σ-algebra identification: the pullback ι^{-1} maps the Baire σ-algebra of $\mathrm{St}(\mathcal{O}_h)$ (generated by clopen sets $[E] = \{u : E \in u\}$ for $E \in \mathcal{O}_h$) to $\mathcal{O}_h = \mathcal{B}$ mod μ.

Measure recovery: CE (Collective Exhaustion), which holds since μ is σ-additive on $\mathcal{O}_h = \mathcal{B}$, forces $\iota_*\mu$ to concentrate on the principal ultrafilters $\iota(X) \subset \mathrm{St}(\mathcal{O}_h)$ (Paper I, CE theorem [4]). So $\iota_*\mu$ is supported on the image $\iota(X)$ and the measure algebras are identified. □

Remark 6.2 (The programme closes)**.** The Stone space $\mathrm{St}(\mathcal{C})$ introduced in Paper I as a tool for deriving σ-additivity from compactness reappears here as the reconstruction of X. The programme begins with an observer whose distinctions generate a Boolean algebra $\mathcal{C}$, and ends by showing that when those distinctions are complete ($\mathcal{O}_h = \mathcal{B}$), the Stone space of $\mathcal{C}$ is the state space of the system they are observing.

The three papers trace the same duality from different angles: Paper I from the Boolean side (Stone duality on the cylinder algebra), Paper II from the spectral side (Koopman operator on L^2), Paper III from the algebraic side (density bridge between the Boolean and L^2 levels).

7 Comparison with Takens

The classical Takens embedding theorem [5] states: for a smooth vector field on a compact d-dimensional manifold M, and for a generic smooth observation function $h : M \to \mathbb{R}$, the delay map

$$\Phi_h^{(k)} : M \to \mathbb{R}^k, \qquad \Phi_h^{(k)}(x) = (h(x), h(Tx), \ldots, h(T^{k-1}x))$$

is an embedding (injective immersion) for $k \geq 2d + 1$.

The reconstruction theorem differs in four respects:

(1) *Regularity.* Takens requires a C^2 diffeomorphism and a C^2 observation. The reconstruction theorem requires only measurability of T and h.

(2) *Window length.* Takens uses a finite window of length $2d+1$, determined by the dimension of M. The reconstruction theorem uses all delays $n \in \mathbb{Z}$ simultaneously. The dimension bound in Takens ensures that the finite-delay algebra already separates points; here the algebra density condition replaces it.

(3) *Embedding type.* Takens produces a diffeomorphic embedding into $\mathbb{R}^{2d+1}$. The reconstruction theorem produces a measure-theoretic embedding into $\mathbb{R}^{\mathbb{Z}}$: an isomorphism of measure algebras, not a topological embedding.

(4) *Genericity.* Takens requires genericity of h (to avoid degenerate observation functions). In the measure-theoretic setting, the corresponding condition is $\mathcal{O}_h = \mathcal{B}$: the observation function generates the full σ-algebra. On a compact manifold, this holds for any h whose level sets $\{h = c\}$ are $(d-1)$-dimensional submanifolds, which is a generic condition on h.

Remark 7.1 (Recovering Takens on manifolds)**.** On a compact d-manifold with a C^2 diffeomorphism T and a generic smooth h, the orbit $\{h \circ T^n : n \in \mathbb{Z}\}$ separates points. By a Stone–Weierstrass argument, the algebra $\mathcal{A}_h$ is uniformly dense in $C(M)$ and hence L^2-dense. So the algebraic density condition of Theorem 4.1(ii) holds, and Φ_h is a measure-theoretic embedding. For Lebesgue measure on M, this is the measure-theoretic content of the Takens theorem.

Remark 7.2 (Lean formalization)**.** The core results of this paper are formalized in Lean 4 using Mathlib [6], in the file `ReconstructionTheorem.lean` (project `QuerySystem`). The following are fully proved with zero sorrys:

- `observableAlgebra`: the σ-algebra $\mathcal{O}_h = \sigma\{h \circ T^n : n \in \mathbb{Z}\}$ (Definition 2.1).
- `observableAlgebra_eq_comap`: $\mathcal{O}_h$ equals the pullback of the product σ-algebra along Φ_h (Lemma 2.3).
- `delayMap_intertwines_shift`: $\Phi_h \circ T = \sigma \circ \Phi_h$, where σ is the unilateral shift on $\mathbb{N} \to \mathbb{R}$ (forward-orbit analogue of Corollary 4.3).
- `cyclic_implies_dense`: if the translates $\{h \circ T^n : n \in \mathbb{N}\}$ span a dense subspace in $L^2(\mu)$, then $L^2(\mathcal{O}_h, \mu)$ is dense in $L^2(\mu)$ (Corollary 5.2).

Two further results carry documented `sorry`s at genuine Mathlib gaps:

- `lpMeas_eq_top_of_ae_eq`: if every $\mathcal{B}$-set has an $\mathcal{O}_h$-a.e.-equal version, then every $f \in L^2(\mu)$ is $\mathcal{O}_h$-a.e.-strongly-measurable. Blocked by the absence of a downward-monotonicity lemma for `AEStronglyMeasurable` under a.e.-equivalent σ-algebras in Mathlib.
- `reconstruction_iff_lpMeas` ($\Leftarrow$): density of $L^2(\mathcal{O}_h, \mu)$ implies every $\mathcal{B}$-set has an $\mathcal{O}_h$-version a.e. (requires an a.e.-convergent subsequence from L^2 convergence; `tendsto_ae_of_tendsto_Lp` not yet in Mathlib).

The formalization is on the `stone-duality-extension` branch of the project repository.

References

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[2] Paul R. Halmos. In general a measure preserving transformation is mixing. *Annals of Mathematics*, 45(4):786–792, 1944.

[3] Patrick S. Mahon. Dynamics from probability: Koopman–Perron duality and the predictive semigroup. Preprint, 2026.

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[5] Floris Takens. Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence, Warwick 1980*, volume 898 of *Lecture Notes in Mathematics*, pages 366–381. Springer, 1981.

[6] The Mathlib Community. Mathlib4: the Lean 4 mathematical library. `https://github.com/leanprover-community/mathlib4`, 2024.